

39. Events: change, input, cut, copy, paste

`change` event

Jab element ne **badalna** khatam kiya.

- **Text input:** focus **khone** par chalta hai (type karte waqt nahi).
- `select`, **checkbox**, **radio**: selection badalte hi turant.

html

```
<input type="text" onchange="alert(this.value)">
```

`input` event

Value badalne par **har baar turant** chalta hai, chahe kaise bhi badli ho: type, mouse se paste, speech recognition.

js

```
input.oninput = function() { result.innerHTML = input.value; };
```

- Input field ke **har** change ko pakadna ho to ye sabse achha.
- Keyboard events ke ulta, jo keys value nahi badalti (jaise arrow keys) unpar nahi chalta.
- `oninput` me `preventDefault()` **kaam nahi karta**, kyunki ye value badalne ke **baad** aata hai.

`cut`, `copy`, `paste`

`ClipboardEvent` class ke events. Data `event.clipboardData` se milta hai, aur `preventDefault()` se action **roka** ja sakta hai.

js

```
input.onpaste = function(event) {  
  alert("paste: " + event.clipboardData.getData('text/plain'));  
  event.preventDefault();  
};  
  
input.ongcut = input.ongcopy = function(event) {  
  alert(event.type + '-' + document.getSelection());  
  event.preventDefault();  
};
```

- `cut/copy` handlers me `clipboardData.getData()` **khaali string** deta hai (data abhi clipboard me gaya nahi), isliye `document.getSelection()` use kiya.
- Sirf text nahi, files bhi copy/paste ho sakti hain (`clipboardData` `DataTransfer` interface hai).
- Naya async API: `navigator.clipboard` (Firefox me support nahi tha).

Safety restrictions

Clipboard OS-level global cheez hai, isliye browsers usse **sirf user ki apni actions** (copy/paste) ke scope me access dete hain.

- `dispatchEvent` se custom clipboard events banana Firefox ke alawa sab me mana hai, aur bane bhi to clipboard access nahi milta.
- `event.clipboardData` ko save karke baad me use nahi kar sakte.
- `navigator.clipboard` kisi bhi context me use ho sakta hai, zarurat par permission maangta hai.

Summary

Event	Kab	Khaas baat
<code>change</code>	Value badli	Text input me focus khone par
<code>input</code>	Text input me har change	Turant, <code>change</code> ke ulta
<code>cut/copy/paste</code>	Cut/copy/paste	Roka ja sakta hai, <code>clipboardData</code> se data

Task

Deposit calculator: amount, percentage, years me koi bhi change turant process ho (`input` event).

```
js
let result = Math.round(initial * (1 + interest) ** years);
```